# Fastmixture

2024-11-14

```
res_dir <- "../results/s05_fastmixture/"
```

This script uses fastMIXTURE (Santander et al., 2024) through the package tidypopgen (Carter and Manica, 2025), and clumppling (ref) through the package tidygenclust (ref) to estimate ancestry proportions of individuals in the global dataset.

## Load gt

```
mad_bor_g1000_sum <-
  gt_load("../data/intermediate/rbind_data/mad_bor_g1000_sum_neo_clean_ld.gt",
    reattach_to = "../data/intermediate/rbind_data/mad_bor_g1000_sum_neo_clean_ld"
  )
mad_bor_g1000_sum <- mad_bor_g1000_sum %>% group_by(population)
```

## Loci and sample count

```
nrow(mad_bor_g1000_sum)
```

```
## [1] 2744
```

```
count_loci(mad_bor_g1000_sum)
```

```
## [1] 198196
```

fastMIXTURE was run on the full, cleaned dataset of 2744 individuals and 198196 SNPs.

```
if (file.exists("../data/intermediate/fastmixture/fastmixture_m.RData")) {
  qmats <-
    readRDS("../data/intermediate/fastmixture/fastmixture_m.RData")
} else {
  tictoc::tic()
  k <- c(2:10)
  qmats <- gt_fastmixture(mad_bor_g1000_sum,
    k,
    n_runs = 3,
    threads = 1,
```

```
    batches = 16,
    seed = c(1, 2, 3),
    no_freqs = TRUE,
    cv = 5,
    cv_tole = 1e-7
  )
  saveRDS(qmats, "../data/intermediate/fastmixture/fastmixture_m.RData")
  time <- tictoc::toc()
}
```

fastMIXTURE was run unsupervised for k values `min(range(unique(qmats$k)))` to 10, with 3 repeats.

```
qmats <- readRDS("../data/intermediate/fastmixture/fastmixture_m.RData")
```

```
# check order
testthat::expect_equal(qmats$id, mad_bor_g1000_sum$id)
testthat::expect_equal(qmats$group, mad_bor_g1000_sum$population)
```

## Reorder gen_tibble

```
mad_bor_g1000_sum$Region <- "Other"
mad_bor_g1000_sum[mad_bor_g1000_sum$population %in% c("Borneo", "Sumba", "Timor"), "Region"] <- "Island
mad_bor_g1000_sum[mad_bor_g1000_sum$population %in% c("BEB", "PJL"), "Region"] <- "South Asia"
mad_bor_g1000_sum[mad_bor_g1000_sum$population %in% c("CHS", "KHV"), "Region"] <- "East Asia"
# Not really happy about grouping all African samples here though...
mad_bor_g1000_sum[mad_bor_g1000_sum$population %in% c("ESN", "GWD", "LWK", "MSL", "YRI"), "Region"] <-
mad_bor_g1000_sum[mad_bor_g1000_sum$population %in% c("Betsimisaraka", "Merina", "Diego", "Sakalava_Tsi
# wharever is left as "Other" set to Africa
mad_bor_g1000_sum[mad_bor_g1000_sum$Region == "Other", "Region"] <- "Africa"

mad_bor_g1000_sum <- mad_bor_g1000_sum %>% arrange(Region, population)
```

## Reorder qmats according to population order in mad_bor_g1000_sum

```
mad_bor_g1000_sum <- mad_bor_g1000_sum %>% filter(.data$id %in% qmats$id)

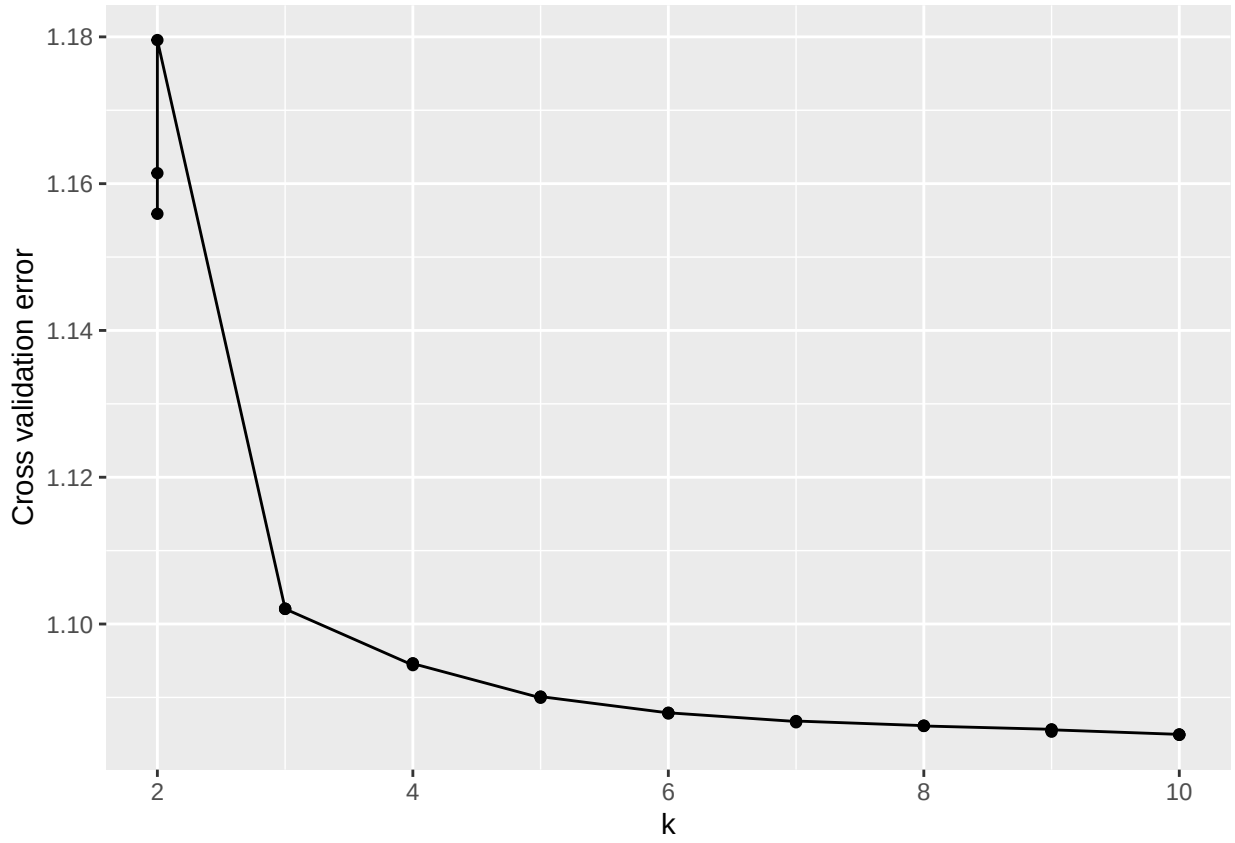
mad_bor_g1000_sum <- mad_bor_g1000_sum[match(qmats$id, mad_bor_g1000_sum$id), ]

region <- mad_bor_g1000_sum$Region
region <- factor(region, levels = unique(mad_bor_g1000_sum$Region))

qmats <- gt_admix_reorder_q(qmats, group = region)
```

## CV

```
autoplot(qmats, type = "cv")
```

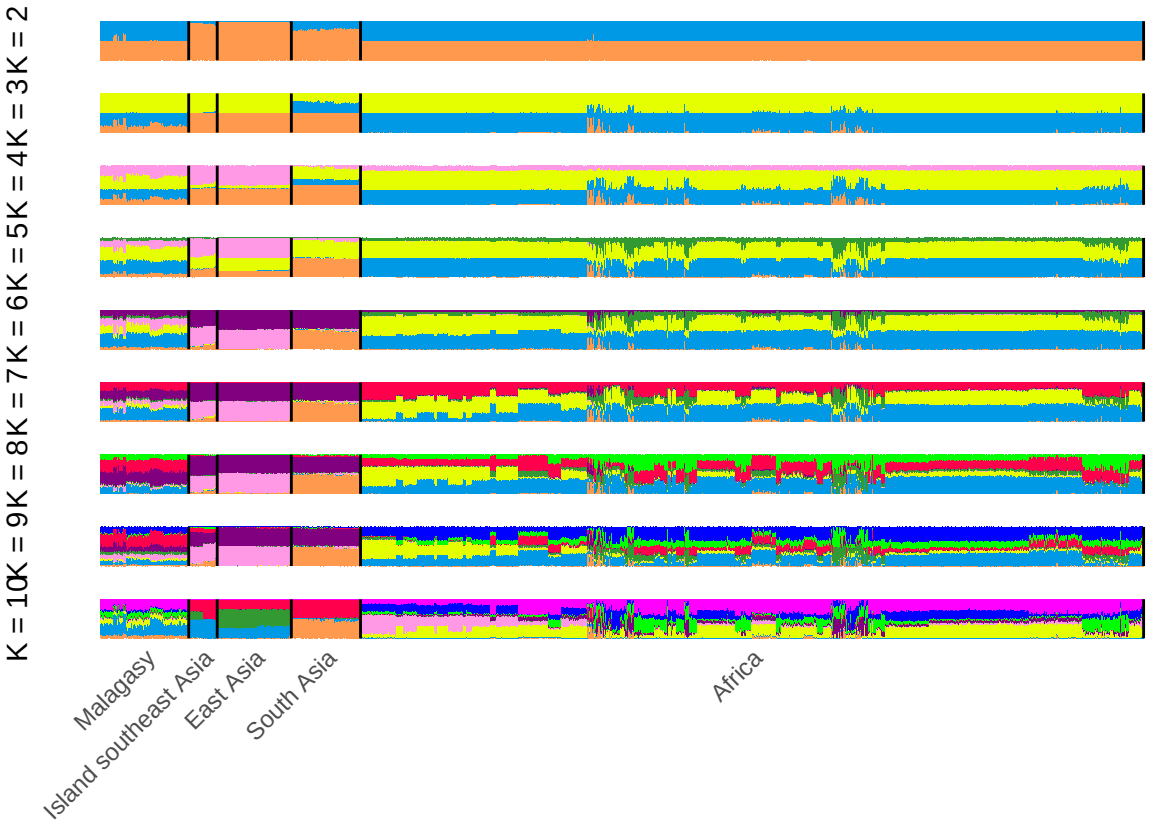



```
ggsave(paste0(res_dir, "fastmixture_cv_error.jpg"),
  width = 7, height = 6, dpi = 300
)
```

## Clumppling

```
clump <- gt_clumppling(input_path = qmats)
```

```
out <- clump %>% autoplot(
  type = "all_modes",
  group = as.character(qmats$group)
)
out
```



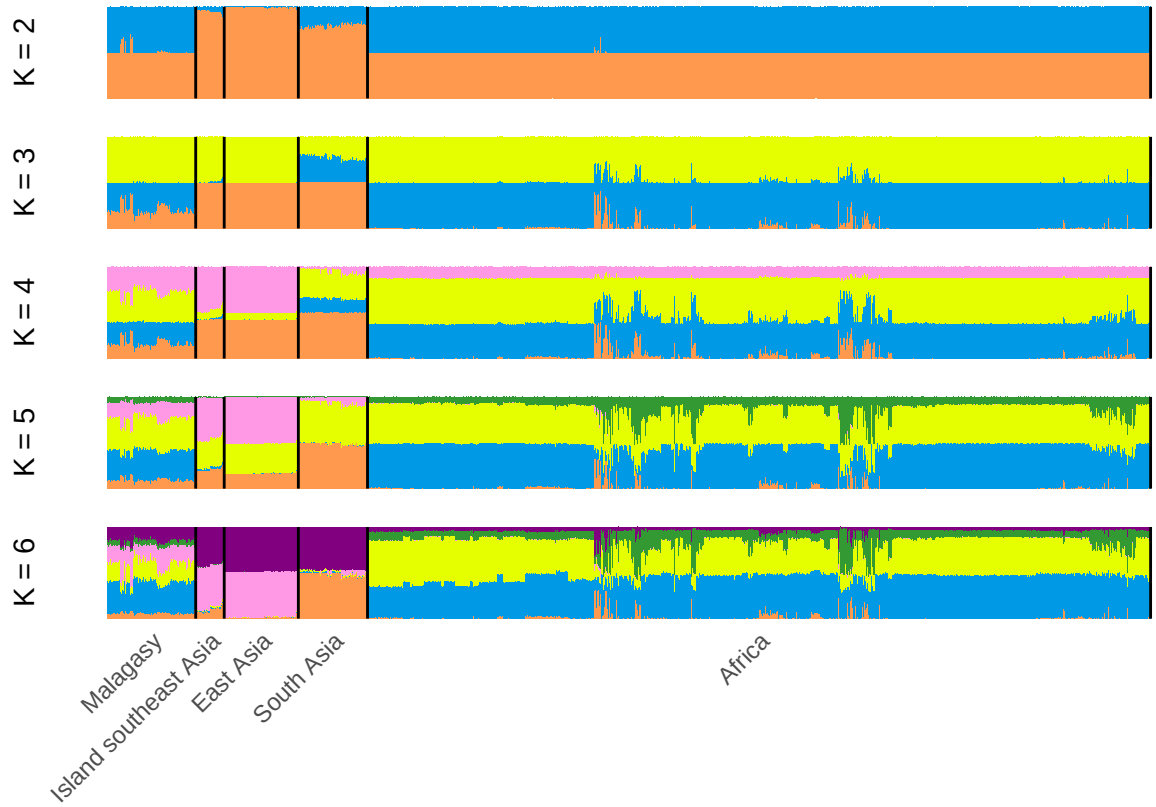

```
ggsave(paste0(res_dir, "clumppling_all_modes.jpg"),
  width = 7, height = 6, dpi = 300
)
```

```
devtools::load_all("/home/EJC199/Git/tidygenclust")
```

```
## i Loading tidygenclust
```

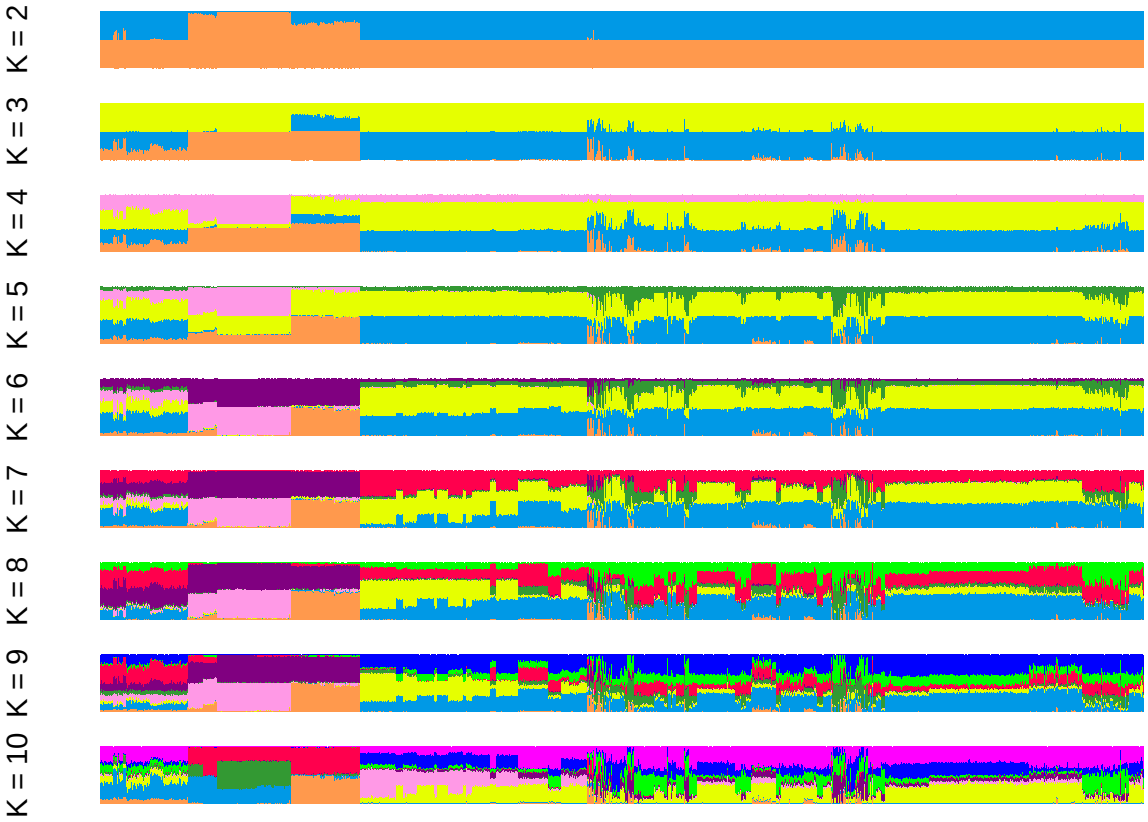
```
# subset according to index
sub_clump <- subset_gt_clumppling(clump, k = 2:6)

# with group labels
sub_clump %>% autoplot(
  type = "all_modes",
  group = as.character(qmats$group)
)
```



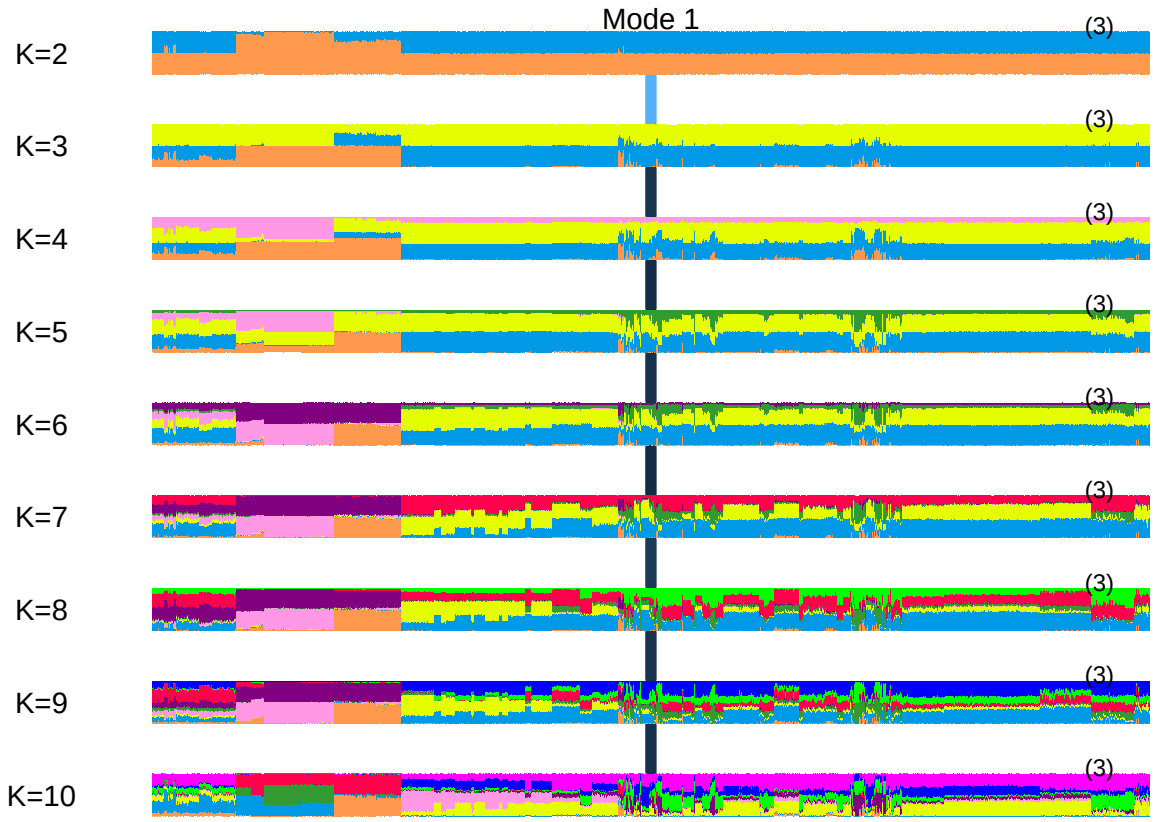

```
ggsave(paste0(res_dir, "clumppling_all_modes_k2_6.jpg"),
  width = 7, height = 6, dpi = 300
)
```

```
clump %>% autoplot(type = "major_modes")
```



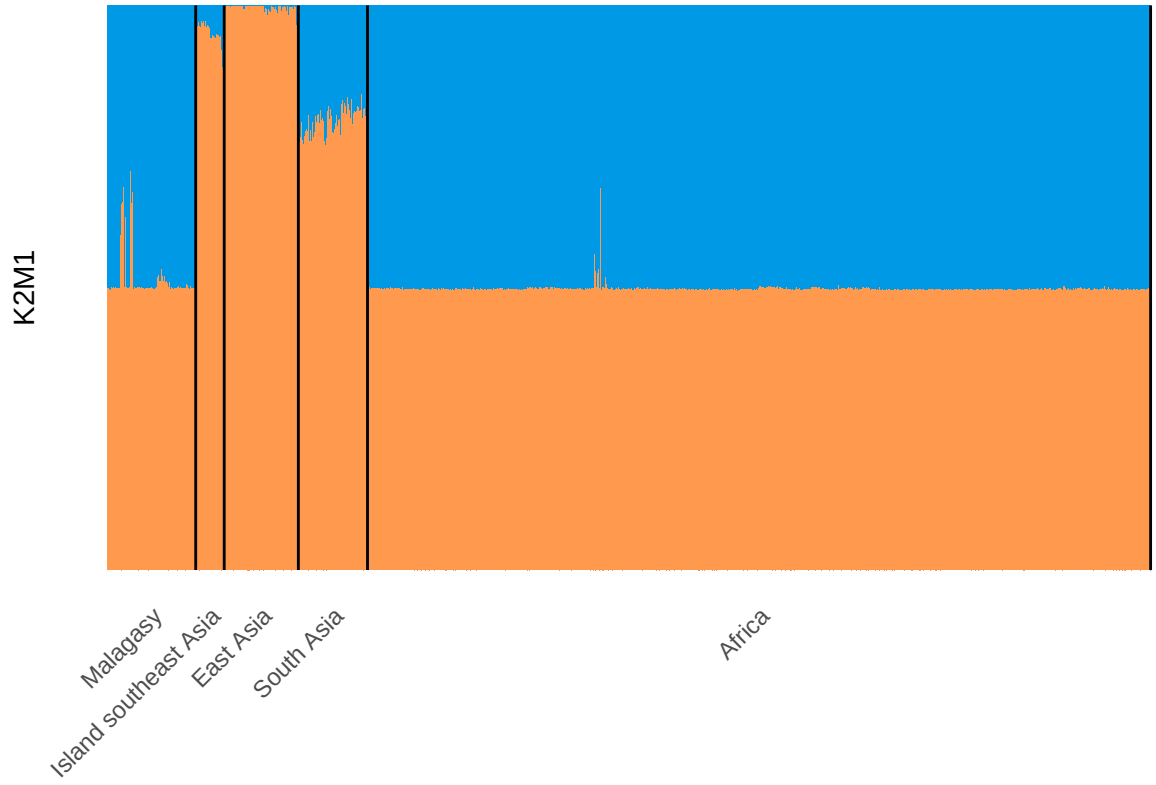

```
ggsave(paste0(res_dir, "clumppling_major_modes.jpg"),
  width = 7, height = 7, dpi = 300
)
```

```
clump_out <- clump %>% autoplot(type = "modes")
clump_out
```




```
ggsave(paste0(res_dir, "clumppling_modes.jpg"),
  width = 7, height = 7, dpi = 300
)
```

```
clump %>% autoplot(
  type = "modes_within_k", k = 2,
  group = as.character(qmats$group)
)
```




```
ggsave(paste0(res_dir, "k2m1.jpg"),
  width = 7, height = 7, dpi = 300
)
```

## Add Malagasy population names

```
mad_bor_g1000_sum$Region <- ifelse(mad_bor_g1000_sum$Region == "Malagasy",
  mad_bor_g1000_sum$population,
  mad_bor_g1000_sum$Region
)

mad_bor_g1000_sum <- mad_bor_g1000_sum %>% arrange(Region, population)
```

## Reorder qmats according to population order in mad_bor_g1000_sum

```
mad_bor_g1000_sum <- mad_bor_g1000_sum %>% filter(.data$id %in% qmats$id)

mad_bor_g1000_sum <- mad_bor_g1000_sum[match(qmats$id, mad_bor_g1000_sum$id), ]

region <- mad_bor_g1000_sum$Region
```

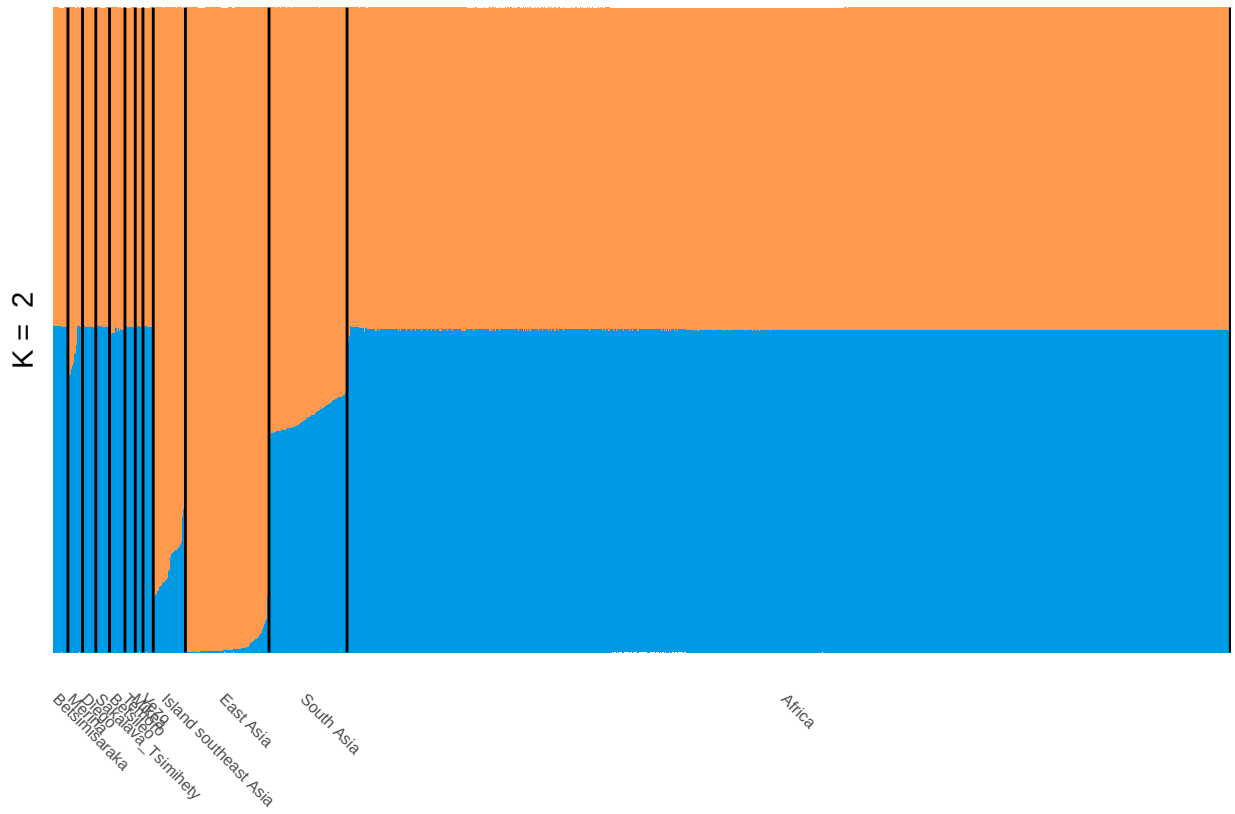
```
region <- factor(region, levels = unique(mad_bor_g1000_sum$Region))

qmats <- gt_admix_reorder_q(qmats, group = region)
```

## Plot

```
mad_bor_g1000_sum <- mad_bor_g1000_sum %>%
  group_by(Region)

autoplot(get_q_matrix(qmats, k = 2, run = 1),
  annotate_group = TRUE,
  arrange_by_group = TRUE,
  arrange_by_indiv = TRUE,
  reorder_within_groups = TRUE
) +
  theme( # Rotate and adjust alignment
    axis.text.x = element_text(angle = -45, hjust = 0, vjust = 1, size = 6)
  )
```




```
ggsave(paste0(res_dir, "fastmixture_autoplot_k2.jpg"),
  width = 7, height = 5, dpi = 300
)
```

```
# rerun clumppling
clump <- gt_clumppling(input_path = qmats)

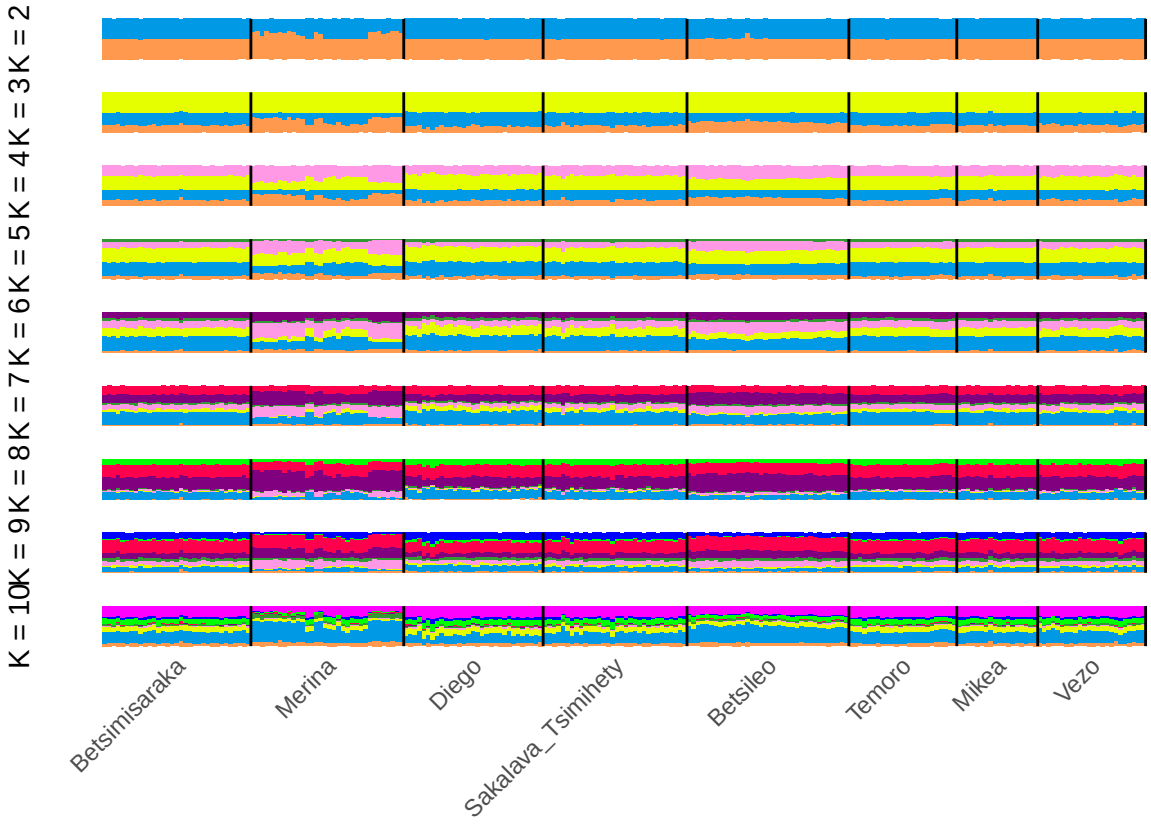
devtools::load_all("/home/EJC199/Git/tidygenclust")
```

```
## i Loading tidygenclust
```

```
# find samples of interest within gt_admix object `qmats`
malagasy_index <- which(qmats$group %in% c(
  "Betsimisaraka", "Merina", "Diego",
  "Sakalava_Tsimihety", "Betsileo",
  "Vezo", "Mikea", "Temoro"
))

# subset according to index
sub_clump <- subset_gt_clumppling(clump, indivs = malagasy_index)

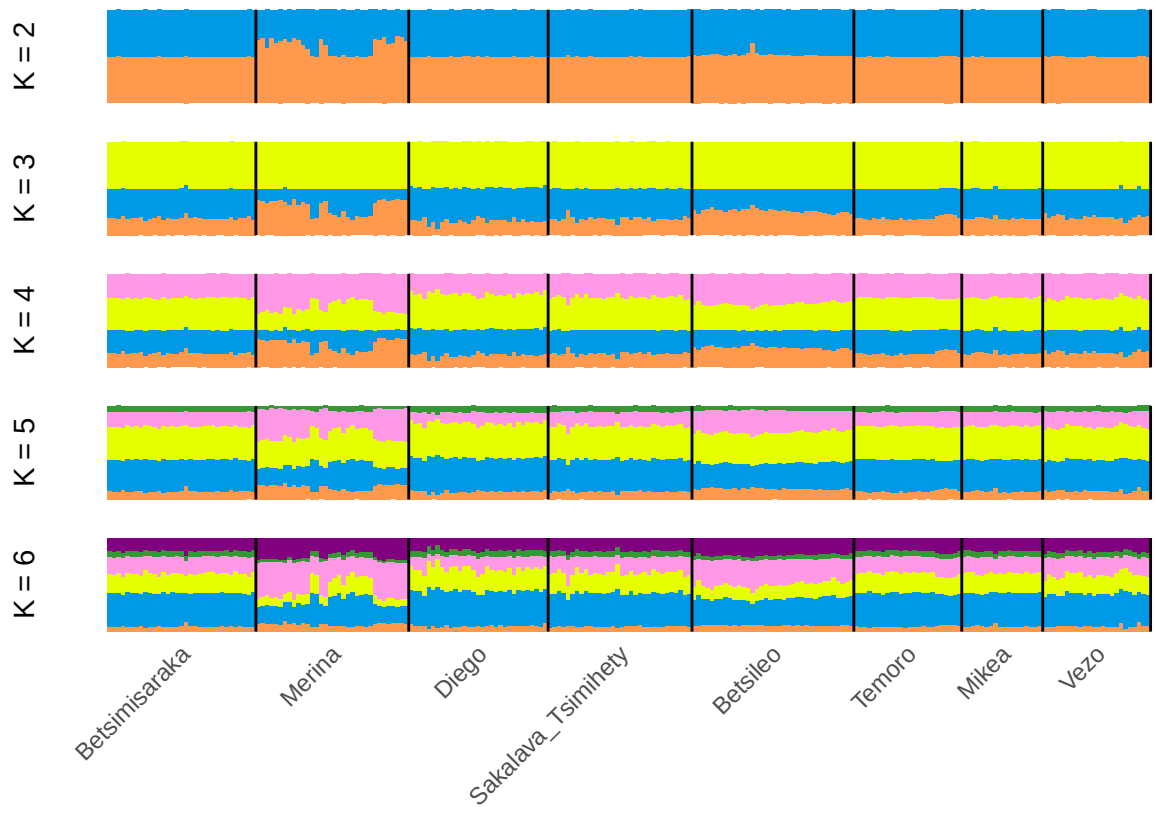
# with group labels
sub_clump %>% autoplot(
  type = "all_modes",
  group = as.character(qmats$group[malagasy_index])
)
```



```
ggsave(paste0(res_dir, "clumppling_all_modes_malagasy_only.jpg"),
  width = 7, height = 6, dpi = 300
)

# subset according to index
sub_clump <- subset_gt_clumppling(clump, k = 2:6, indivs = malagasy_index)

# with group labels
sub_clump %>% autoplot(
  type = "all_modes",
  group = as.character(qmats$group[malagasy_index])
)
```




```
ggsave(paste0(res_dir, "clumppling_all_modes_malagasy_only_k2_6.jpg"),
  width = 7, height = 6, dpi = 300
)
```

## Check which ancestry components each .Q represents

```
augmented <- augment(get_q_matrix(qmats, k = 2, run = 1),
  data = mad_bor_g1000_sum
)
```

```r
# filter augmented to $population CHS and print head
augmented_chs <- augmented %>% filter(population == "CHS")
head(augmented_chs) # .Q1 representing Asian ancestry
```

```
## # A tibble: 6 x 9
## # Groups:   Region [1]
##   id      population sex    genotypes Region    .rownames   .Q1      .Q2 group
##   <chr>   <chr>      <chr> <vctr_SNP> <chr>         <int> <dbl>   <dbl> <int>
## 1 HG00403 CHS        <NA>   [1,1,...] East Asia       308 0.997 0.00320    10
## 2 HG00404 CHS        <NA>   [1,0,...] East Asia       309 0.998 0.00192    10
## 3 HG00406 CHS        <NA>   [2,1,...] East Asia       310 0.998 0.00176    10
## 4 HG00407 CHS        <NA>   [2,1,...] East Asia       311 0.996 0.00357    10
## 5 HG00409 CHS        <NA>   [1,0,...] East Asia       312 0.999 0.00102    10
## 6 HG00410 CHS        <NA>   [1,2,...] East Asia       313 0.996 0.00443    10
```

```r
augmented_gwd <- augmented %>% filter(population == "GWD")
head(augmented_gwd)
```

```
## # A tibble: 6 x 9
## # Groups:   Region [1]
##   id      population sex    genotypes Region .rownames   .Q1   .Q2 group
##   <chr>   <chr>      <chr> <vctr_SNP> <chr>      <int> <dbl> <dbl> <int>
## 1 HG02461 GWD        <NA>   [0,0,...] Africa       685 0.500 0.500    12
## 2 HG02462 GWD        <NA>   [2,1,...] Africa       686 0.501 0.499    12
## 3 HG02464 GWD        <NA>   [2,0,...] Africa       687 0.500 0.500    12
## 4 HG02465 GWD        <NA>   [1,0,...] Africa       688 0.500 0.500    12
## 5 HG02561 GWD        <NA>   [1,0,...] Africa       689 0.500 0.500    12
## 6 HG02562 GWD        <NA>   [0,0,...] Africa       690 0.500 0.500    12
```

---

## Recording R environment

Part of ensuring our workflow is reproducible is to keep track of the R environment. We will automatically save our R environment into a file called `r_environment.json`.